HyperProve: Answer-Guided Hypergraph Expansion for Multi-Hop Question Answering

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Introduction & Motivation

- **Multi-hop QA** needs evidence connected through intermediate (*bridge*) entities, relations and constraints.
- **The Bottleneck:** Retrieval scores facts against the *original* question, but the next useful fact only becomes relevant after an earlier hop is resolved ⇒ it is missed, or retrieval drifts to a look-alike distractor.
- **Prior work:** graph RAG keeps structure but uses a fixed frontier; iterative RAG rewrites queries but carries only *text*, losing the supporting evidence.
- **HyperProve:** carry the *answer + its supporting atomic facts* as the structural frontier of the next hop.

Methodology

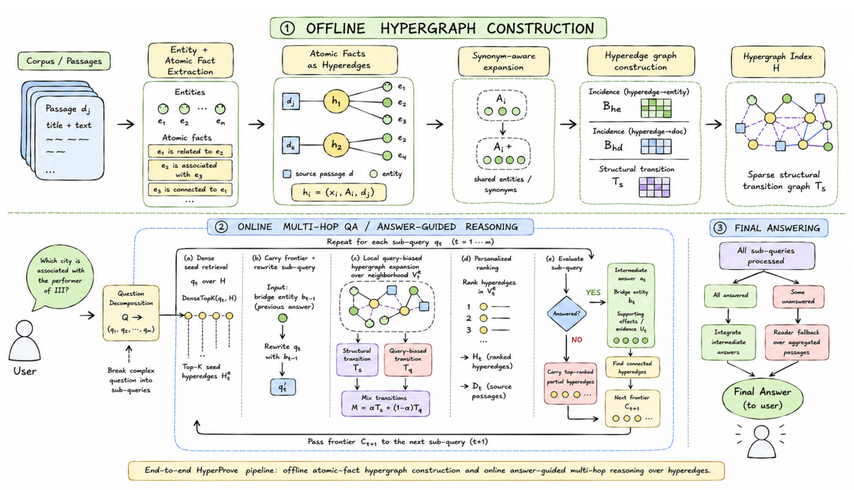


Figure 1: Overall pipeline: offline atomic-fact hypergraph construction + online answer-guided multi-hop reasoning.

Atomic-Fact Hypergraph (offline)

- Each passage is split into self-contained atomic facts; a fact is a **hyperedge** linking all its participants and its source passage: $h_i = (x_i,\ \mathcal{A}_i,\ d_{\sigma(i)})$.
- Synonym-aware entities ($\cos > 0.8$). Facts are linked if they share an entity or a passage; inverse degrees down-weight hubs:

$$W_s = \underbrace{B_{he}D_e^{-1}B_{he}^\top}_{\text{shared entity}} + \underbrace{B_{hd}D_d^{-1}B_{hd}^\top}_{\text{same passage}}, \qquad T_s = \mathrm{RowNorm}\big(W_s - \mathrm{diag}(W_s)\big)$$

Query-biased Hyperedge Retrieval

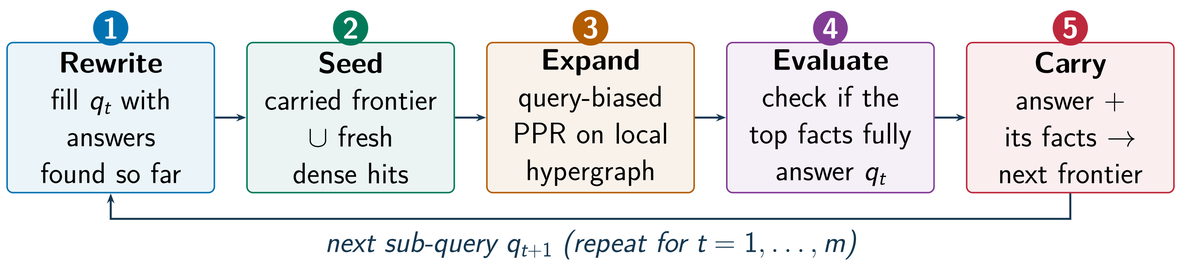
- In a local graph $G_t^\star$ around the seeds, neighbors are re-weighted by similarity to the rewritten sub-query q_t' (kept only if $c_{t,k} > \theta$):

$$c_{t,k} = \cos\big(\phi(q_t'), \phi(x_k)\big), \qquad T_{q,t}^\star[i,k] = \frac{\exp(c_{t,k}/\tau)}{\sum_{\ell \in \mathcal{N}_{t,\theta}(i)} \exp(c_{t,\ell}/\tau)}$$

- Mix structure and query relevance, then rank facts with Personalized PageRank:

$$M_t = \alpha T_{s,t}^\star + (1-\alpha) T_{q,t}^\star, \qquad \pi_t^{(r+1)} = (1-\gamma)\, u_t^\star + \gamma M_t^\top \pi_t^{(r)}$$

Answer-Guided Hyperedge Reasoning (Online)



Algorithm 1: HyperProve: answer-guided reasoning over hyperedge index $\mathcal{H}$

$(q_1, \ldots, q_m) \leftarrow \textsc{Decompose}(Q)$; $\mathcal{R}, \mathcal{C}_1 \leftarrow \emptyset$; ▷ answers so far, carried frontier

for $t = 1$ ***to*** m **do**

- ❶ $q_t' \leftarrow \textsc{Rewrite}(q_t, \mathcal{R})$; ▷ "… named after *Peter*?"
- ❷ $S_t \leftarrow \textsc{TopK}(\mathcal{C}_t, 10) \cup \textsc{DenseTopK}(q_t', \mathcal{H}, 10)$;
- ❸ $(\mathcal{H}_t, \pi_t) \leftarrow \textsc{LocalPPR}(S_t, q_t')$; ▷ rank facts with M_t
- ❹ $(\text{ok}, a_t, \mathcal{U}_t) \leftarrow \textsc{Evaluate}(q_t, \mathcal{H}_t)$; ▷ $\mathcal{U}_t$: supporting facts
- ❺ **if** ok **then** $\mathcal{R}$ += (q_t, a_t); $\mathcal{C}_{t+1} \leftarrow \textsc{Hyperedges}\big(a_t \cup \textsc{Entities}(\mathcal{U}_t)\big)$;
- **else** $\mathcal{C}_{t+1} \leftarrow \textsc{TopK}(\mathcal{H}_t, \pi_t, 10)$ ▷ keep partial evidence;

end

return $\textsc{Integrate}(Q, \mathcal{R})$ *if all hops answered, else* $\textsc{Reader}(Q,$ *top-10 passages*$)$

Main Results

- **Benchmarks:** MuSiQue, HotpotQA and 2WikiMultiHopQA (1,000 questions each).
- **Metrics:** LLM-as-a-judge Accuracy, Exact Match (EM) and token-level F1; mean of three runs.
- **Controlled setup:** every method uses BGE-M3 as retriever and GPT-OSS-20B as backbone. Baselines cover dense, structured (graph) and iterative retrieval.

Table 1: Accuracy (Acc.), Exact Match (EM) and F1 on three multi-hop QA benchmarks.

Methods	MuSiQue			HotpotQA			2WikiMultiHopQA		
	Acc.	EM	F1	Acc.	EM	F1	Acc.	EM	F1
Contriever	47.80	22.30	32.10	55.55	44.40	57.50	69.60	38.50	43.70
BGE-M3	51.50	27.80	39.70	68.12	47.80	61.80	72.18	48.20	54.50
HippoRAG2	52.00	37.20	48.60	81.70	62.70	75.50	86.40	65.00	71.00
PropRAG	50.90	29.30	46.73	85.80	57.80	73.79	77.50	58.00	68.78
HyperGraphRAG	55.13	42.20	53.80	77.89	63.90	73.80	77.72	68.30	71.30
HGRAG	58.60	41.19	55.04	80.10	54.90	69.43	85.00	64.10	74.10
IRCoT	56.67	34.10	47.60	85.23	55.70	71.20	83.65	60.70	74.30
T2RAG	54.67	34.30	45.60	81.33	54.20	67.30	83.75	66.70	74.40
HopRAG	67.18	42.40	54.90	85.51	61.10	68.26	86.66	62.00	76.06
ChainRAG	60.01	41.20	54.15	85.46	60.70	74.66	82.48	63.30	74.39
HyperProve	**71.60**	**42.83**	**59.23**	**91.30**	**63.92**	**76.64**	**91.40**	**69.70**	**80.01**

Row groups: dense / structured (graph) / iterative retrieval. Underline: strongest baseline.

- **Best on every metric** across all three benchmarks. Accuracy gains over the strongest baseline: **+4.42** (MuSiQue, vs. HopRAG), **+5.50** (HotpotQA, vs. PropRAG), **+4.74** (2Wiki, vs. HopRAG).
- Gains are consistent on EM and F1 as well, not only on LLM-judged accuracy.

Ablation Study

Effect of Retrieval Components

Variant (MuSiQue Acc.)	Overall	2-hop	3-hop	4-hop
w/o decomposition	51.28	58.11	49.05	33.73
w/o query transition $T_q^\star$	58.17	65.20	58.60	35.40
w/o fresh dense seed	61.30	71.24	59.18	34.34
w/o structural transition T_s	62.13	67.02	64.65	42.06
w/o bridge entity	64.50	69.11	67.41	44.58
HyperProve (Full)	**71.60**	**78.00**	**72.96**	**49.40**

*Every component helps; decomposition and the query-biased transition matter most, and the gap widens on 4-hop questions.

Evidence Chain Faithfulness

Method	MuSiQue	HotpotQA	2Wiki	Gain
HopRAG	66.63	82.82	84.17	+25.98
ChainRAG	67.42	89.45	75.85	+26.35
HyperProve	**82.31**	**96.20**	**98.15**	**+41.24**

Recall@10 of gold passages; *Gain*: avg. coverage gain after hop 1.

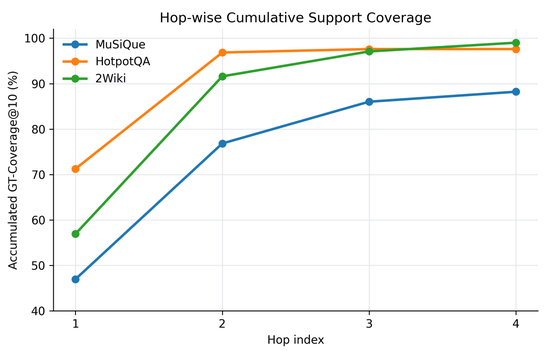


Figure 2: Later hops recover missing support evidence.

Online Efficiency (MuSiQue)

Method	Acc.	Tokens	Lat.
BGE-M3	51.50	1.7M	4.1K
IRCoT	56.67	4.4M	3.8K
T2RAG	54.67	2.4M	9.6K
HopRAG	67.18	36.1M	45.2K
ChainRAG	60.01	15.4M	37.0K
HyperProve	**71.60**	**9.6M**	**31.9K**

1,000 questions; latency in seconds; judge cost excluded. HyperProve uses **3.8×** **fewer tokens** than HopRAG while being **+4.42** more accurate.

Conclusion

- HyperProve carries each intermediate answer *together with* its supporting atomic-fact hyperedges as the retrieval state for the next hop.
- Query-biased expansion over a synonym-aware hypergraph yields the best accuracy, EM and F1 on three multi-hop QA benchmarks, with more faithful evidence chains at moderate cost.

Acknowledgments

This research was funded by Vingroup Innovation Foundation (VINIF) under project code VINIF.2026.DA117.